



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

B. Sc. CHEMISTRY

Programme	B. Sc. Chemistry				
Course Title	PHYSICAL CHEMISTRY – III- CHEMICAL AND PHASE EQUILIBRIA, ELECTROCHEMISTRY AND PHOTOCHEMISTRY				
Type of Course	MAJOR				
Semester	VI				
Academic Level	300 – 399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	CHE2CJ102 and CHE4CJ205. The students must be familiar with the basic concepts in electrochemistry and must have taken NCERT plus-two chemistry course similar to https://onlinecourses.swayam2.ac.in/nce19_sc17/preview				
Course Summary	The chemical reactions tend to reach a state of dynamic equilibrium, i. e., the forward and reverse reactions occur at equal rates. This course introduces the underlying thermodynamic principles that can explain this state of equilibrium. Similarly, thermodynamics of phase transitions and phase equilibria are also explained as the second module. In the third module, electrochemical processes are explained which also involves thermodynamic concepts. In the fourth module, interaction of light with molecules and corresponding chemical as well as physical processes are explained. The final module consists of practical experiments related to these four important topics of chemistry.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	To comprehend the concepts of law of mass action and chemical equilibria	U	F	Assignment/Test

CO2	To understand various phase transitions, construction of phase diagram and its importance in industry	Ap	P	Assignment/Quiz
CO3	To apply the basic concepts of electrochemistry in constructing electrochemical cells	Ap	P	Assignment/Quiz
CO4	To evaluate the pH of buffers, conduct potentiometric titrations and conductivity measurements	An	P	Lab work/Quiz
CO5	To know the theory and working of new generation electrochemical power storage systems	U	C	Assignment/Test
CO6	To understand the photochemical principles and to apply the unknown concentration a given sample solution using colorimetry	Ap	P	Discussion/Lab work
* - Remember (R), Understand (U), Apply (Ap), Analyze (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (45+30)	Marks
I	CHEMICAL EQUILIBRIUM		10	21
	1	Law of mass action, thermodynamic derivation of law of chemical equilibrium.	1	
	2	Relation between Gibbs free energy of reaction and reaction quotient.	1	
	3	Equilibrium constants and their quantitative dependence on temperature, pressure and thermodynamic derivation of relations between the various equilibrium constants K_p , K_c and K_x (using chemical potential).	3	
	4	Van't Hoff's equation - Le Chatelier principle and its application. Homogeneous and heterogeneous equilibria. Clausius-Clapeyron equation - Applications to solid-liquid, Liquid-vapour, Solid- vapour equilibria	3	

	5	Ionic equilibrium: Ionic product of water, pH and pOH, Buffer action, pH of buffer solutions	2	
II	PHASE EQUILIBRIUM		15	33
	6	Concept of phase, Components and Degrees of freedom, Gibbs Phase Rule – Thermodynamic derivation.	3	
	7	Phase diagram for one component systems – Water, CO ₂	2	
	8	Two Component Systems - Phase diagrams for systems involving eutectic – KI/Ice – Freezing mixtures; Pattinson's process. Two-component systems with formation of congruent melting compound-MgZn ₂ . Fractional distillation. Azeotropes.	4	
	9	Partial miscibility of liquids, CST, Miscible pairs, steam distillation Three component systems, water-chloroform-acetic acid system, triangular plots, tie lines	4	
	10	Nernst distribution law: its derivation and applications in solvent extraction.	2	
III	ELECTROCHEMISTRY		15	33
	11	Electrochemical cells. Origin of electrode potentials-half cell potential-standard hydrogen electrode, reference electrodes	2	
	13	Electrochemical series, applications	1	
	14	Cell potential, Nernst equation for electrode and cell potentials. Nernst equation for electrode potential and EMF of a cell	2	
	15	Relationship between free energy and electrical energy. Gibbs Helmholtz equation to galvanic cells.	2	
	16	Concentration cells: Concentration cells with and without transference – Liquid junction potential (LJP).	2	
	17	Application of EMF measurements- Solubility of sparingly soluble salt, determination of pH, Potentiometric titrations	2	
	18	Electrochemical power storage and sources- Primary batteries- Dry cell, Storage batteries- Lead acid battery, Ni-Cd battery, Li-ion battery (basic idea only), Fuel cells- Hydrogen-Oxygen fuel cell, Electrochemical capacitors and supercapacitors (basic idea only)	3	
	19	Corrosion of metals- electrochemical theory- Methods to prevent corrosion	1	

IV	PHOTOCHEMISTRY		5	11
	20	Interaction of light with matter and Beer-Lambert's law, Photochemical process and quantum yield	1	
	21	Photochemical hydrogen-chlorine and hydrogen-bromine reactions-Reasons for high and low quantum yield	2	
	22	Photophysical processes: Jablonski diagram – Fluorescence – Phosphorescence. Non-radiative processes: Internal conversion, inter system crossing and vibrational relaxation. Quenching of fluorescence – Stern – Volmer equation;	1	
	22	Photosensitization - Chemiluminescence. Bioluminescence, thermoluminescence.	1	
V	PHYSICAL CHEMISTRY- PRACTICALS-3		30	
		A minimum of 5 practical experiments out of which at least one each from sections 1, 2 and 3 must be performed and reported. For plots/graphs, suitable softwares may be used and printed hard copies may be presented. Practical records may be in handwritten or computer-printed form. Section 1 1. Construction of phase diagram & determination of eutectic composition and eutectic temperature: <i>Naphthalene-biphenyl</i> system, <i>Naphthalene-diphenylamine</i> system, <i>Biphenyl– diphenylamine</i> system. Section 2 2. Influence of KCl/NaCl impurity on miscibility temperature of phenol–water system and determination of concentration of given KCl/NaCl solution. Section 3 3. Verification of Beer-Lambert law for KMnO_4 and $\text{K}_2\text{Cr}_2\text{O}_7$ & determination of concentration of the given solution. 4. Colorimetric Estimation of iron (in ferric alum solution) 5. Colorimetric Estimation of chromium (in potassium dichromate solution) Section 4 6. Conductometric titration of strong acid and strong base. 7. Potentiometric titration of strong acid and strong base. 8. Preparation of Acidic Buffer and recording pH using pH meter.	3 3 3 3 3 3 3	

		9. Open ended 10. Open ended	3 3	
		References 1. P. W. Atkins, J. de Paula, <i>Atkin's Physical Chemistry</i> 8th Ed., Oxford University Press, 2006. 2. P. W. Atkins, J. de Paula <i>The Elements of Physical Chemistry</i> 7 th Edn., Oxford University Press, Oxford, 2016. 3. B.R. Puri, L.R. Sharma, M.S. Pathania, <i>Principles of Physical Chemistry</i> , 46 th Edn., Vishal Publishing Company, New Delhi, 2013. Further Reading (Module II) 4. S. Glasstone and D H Lewis, <i>Elements of Physical Chemistry</i> , 2 nd Edn., Macmillan & Company, UK, 1962. 5. D. A. McQuarrie, J. D. Simon, <i>Physical Chemistry: A Molecular Approach</i> , University Science Books: Sausalito, CA; 1997. (Module III) 6. S. Glasstone, <i>An Introduction to Electrochemistry</i> . East-West Press Pvt. Ltd., New delhi, 2007. 7. Praveen Tyagi, <i>Electrochemistry</i> , Discovery Publishing House, 2006. (Module III) 8. K.K. Rohatgi-Mukherjee, <i>Fundamentals of Photochemistry</i> , New Age International, 1978.		

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO 4	PS O 5	PS O 6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	3	2	-	-	3	2	3	2	2	-	2	-	1
CO 2	3	2	-	-	3	2	3	2	1	-	2	-	1
CO 3	3	2	-	-	3	2	3	2	1	-	2	-	1
CO 4	3	1	2	2	3	3	3	2	1	1	1	1	1
CO 5	3	2	-	1	3	3	3	2	1	-	3	-	1
CO 6	3	-	2	3	3	3	3	2	1	2	3	2	1

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Discussion / Seminar
- Midterm Exam
- Practical exam (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment /viva	Practical skill evaluation	End Semester Examinations
CO 1		✓		✓
CO 2		✓		✓
CO 3	✓			✓
CO 4	✓			✓
CO 5	✓	✓		✓
CO 6	✓	✓	✓	✓